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Explanations of Large Language Models Explain Language Representations in the Brain

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Abstract

Large language models (LLMs) not only exhibit human-like performance but also share computational principles with the brain's language processing mechanisms. While prior research has focused on mapping LLMs' internal representations to neural activity, we propose a novel approach using explainable AI (XAI) to strengthen this link. Applying attribution methods, we quantify the influence of preceding words on LLMs' next-word predictions and use these explanations to predict fMRI data from participants listening to narratives. We find that attribution methods robustly predict brain activity across the language network, revealing a hierarchical pattern: explanations from early layers align with the brain's initial language processing stages, while later layers correspond to more advanced stages. Additionally, layers with greater influence on next-word prediction—reflected in higher attribution scores—demonstrate stronger brain alignment. These results underscore XAI's potential for exploring the neural basis of language and suggest brain alignment for assessing the biological plausibility of explanation methods.

Keywords: Cognitive computational Neuroscience; Language Models; Explainable AI; Encoding Model; fMRI

1 Introduction

Recent progress in deep learning has led to the development of autoregressive language models that excel in capturing language structures and performing human-like capabilities in various linguistic tasks [1–4]. This progress poses a critical question at the intersection of artificial intelligence (AI) and neuroscience: Are the observed similarities between language models and human cognition merely superficial, or do they stem from shared underlying mechanisms?

Previous studies have demonstrated significant alignment between large language models (LLMs) and brain activity, primarily through linear mappings between neural responses and LLM internal representations, such as activations, attention heads, and layer transformations [5–8]. This relationship has been explored across multiple dimensions, including model layers, architectures, training settings, and linguistic performance [5, 6, 9, 10]. Notably, transformer models consistently outperform recurrent

and embedding-based models in brain alignment tasks, displaying brain-like patterns tied closely to their proficiency in next-word prediction tasks [5, 6].

Building on this foundation, subsequent research has leveraged LLM-brain alignment to deepen our understanding of both systems. For example, LLM-based encoders have provided insights into key aspects of neural processing, such as predictive processing, semantic selectivity, and meaning composition in the brain during naturalistic language processing [11–13]. Conversely, these approaches have been employed to evaluate and refine LLMs themselves [9, 14, 15].

Despite these advances, several critical questions remain unresolved: (i) Do the observed similarities between LLMs and the brain arise from shared pathways between these two systems, or are they simply artifacts of high-dimensional feature spaces [16, 17]? (ii) Beyond internal representations, are there alternative frameworks that can more effectively capture the alignment between LLMs and neural language processing? Addressing these questions requires moving beyond treating both systems as "black boxes" [18], and instead probing their underlying mechanisms in a more interpretable way.

To address these questions, we leverage explainable AI (XAI) techniques, specifically attribution methods, which quantify how much each preceding word contributes to a model's evolving representation and next-word prediction. Understanding how meaning is constructed across words is also a critical goal for neurocomputational theories of language processing [8]. If the brain employs a strategy similar to LLMs in weighting prior linguistic inputs, attribution-based explanations could provide a novel framework for modeling the neural mechanisms underlying this process in the brain (Fig. 1a).

In this study, we applied four types of attribution methods to three LLMs and constructed feature representations to model fMRI activity recorded during naturalistic story listening. We then evaluated the predictive power of these features in capturing brain activity, comparing them to traditional internal representations such as model activations and attention weights (Fig. 1b, c). To investigate how attribution-based explanations vary across different model layers, we employed layer conductance to generate feature representations for individual layers and assessed their alignment with specific brain regions.

Our findings reveal the strong predictive capability of attribution methods across a wide network of language-related brain areas, consistently observed across participants. Notably, attribution methods outperform internal representations in predicting brain activity in early language-processing regions. Furthermore, we uncovered a striking hierarchical correspondence between LLMs and the brain, suggesting shared mechanisms for integrating contextual information across processing stages. These results position attribution methods as a powerful tool for bridging LLMs and the brain, offering new insights into the computational principles underlying language processing in both artificial and biological systems.

2 Results

Are LLM-derived explanations effective in explaining language processing in the brain? We explore this question through an encoding modeling approach that predicts brain activity based on explanations derived from attribution methods. Specifically, we focus on explanations that clarify why an LLM makes a particular prediction, providing a more transparent window into the model's decision-making process.

We utilized brain data from a subset of the Narratives fMRI dataset [19], which includes 147 participants who listened to four distinct audio stories. We then processed the same stories using LLMs with a sliding window approach, where each model separately predicted the next word based on the preceding context. Attribution methods—a class of XAI techniques—were applied to quantitatively assess how specific words influence the model's decision-making (i.e., predicting the next word). These methods assign importance scores to each word, reflecting its impact on the LLM's predictions. The scores were subsequently organized into a matrix to create a feature space (Fig. 1b).

In the encoding modeling step, the feature spaces derived from each LLM were independently used to predict participants' brain responses. Predictions were made using a voxel-wise linear ridge regression model, tailored to each individual's brain data. Prediction accuracy was evaluated using five-fold cross-validation, with the "brain score" metric as the evaluation criterion [20, 21]. This metric

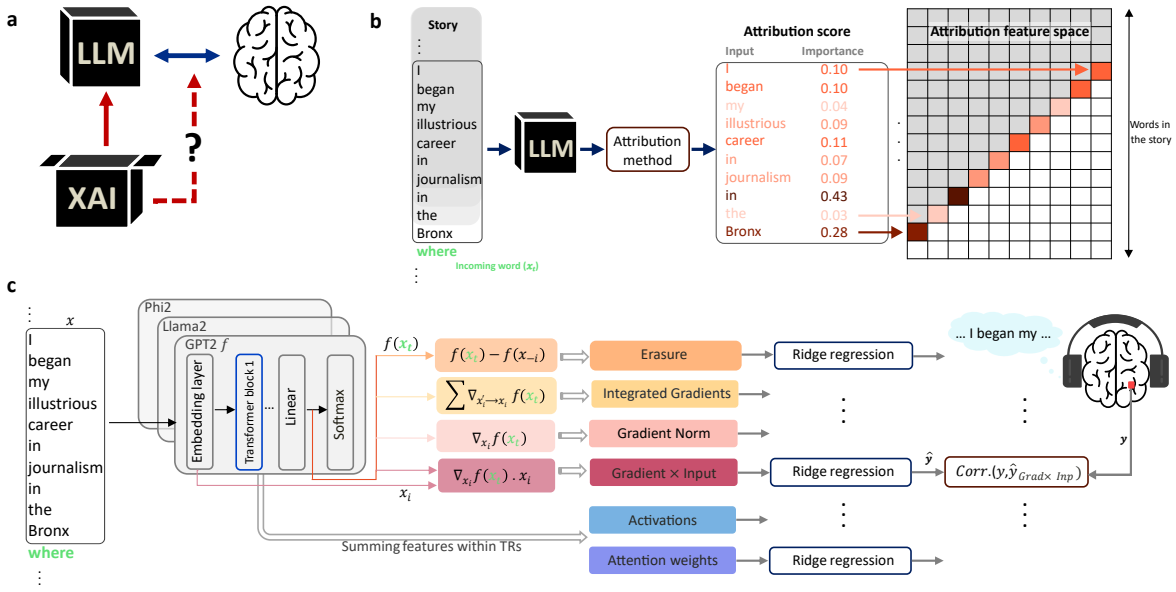

Fig. 1 Approach. a, Language processing in the brain and LLMs shows striking parallels. Applying XAI methods, this study investigates how XAI can reveal insights into both systems and their interrelationship. **b,** To test this hypothesis, we used attribution methods as representatives of XAI. Attribution methods reveal how much each previous word in a context influences the model's decision about the next incoming word. Using a sliding window approach, we applied these methods to continuous stories. Since the sliding windows overlap, each word in the story is processed within different windows, resulting in multiple importance scores being assigned to a single word. These scores were stored in a vector for each word, forming an attribution feature space. **c,** We used ridge regression to quantify how well attribution-based explanations predict fMRI brain activity recorded while participants listened to stories. Prediction accuracy was assessed with a brain score, defined as the Pearson correlation between the predicted and actual brain responses on held-out data. We applied this approach to attribution feature spaces generated by four distinct attribution methods (Erasure, Integrated Gradients, Gradient Norm, and Gradient × Input) across three LLMs (GPT-2, Llama 2, and Phi-2), using four stories. Each feature space was independently tested to predict the fMRI activity of participants for its corresponding story. In total, 147 participants were analyzed across all stories. Additionally, we compared these attribution-based feature spaces to widely used internal LLM representations, namely activations and attention weights, using the same stories, LLMs, and participants. In the figure: x represents the input sequence, x_t denotes the target word (incoming word), and x' is the baseline input. f represents the LLM function, while y is the recorded brain response (BOLD signal) from a voxel, and $\hat{y}$ is the predicted response. *Corr.* refers to the Pearson correlation used for evaluation. The attribution methods formulas are shown in full in the Methods.

computes the Pearson correlation between the predicted and observed fMRI responses in held-out data (Fig. 1c).

2.1 Explanations

In this study, we leverage feature attribution methods, a key class of XAI techniques, to gain insights into how LLMs generate predictions. These methods quantify the contribution of each token in a given context to the model's next-word prediction, providing a fine-grained view of the model's internal decision-making process. Feature attribution techniques have proven valuable across various applications, including improving model interpretability [22], identifying biases [23], and diagnosing model errors [24]. In this study, we employ four distinct attribution methods, each of which is described in detail below.

Gradient Norm: This method computes the gradient of the model's output with respect to the input token embeddings and returns the L1 norm of the resulting gradient as the importance score for each token [25, 26]. The L1 norm ensures that the importance scores reflect the magnitude of the gradient, providing a straightforward measure of each token's influence on the model's prediction.

Gradient × Input: A key consideration in computing importance scores is the role of input token embeddings. The Gradient × Input method [27] addresses this by multiplying the gradient of the